

Bulan 5: Advanced DRF & Production Concepts

Minggu 1: Advanced Authentication & Security

Hari 1: JWT Authentication

Topik: JSON Web Tokens, djangorestframework-simplejwt, access & refresh tokens

Proyek: JWTAuthAPI

API dengan JWT authentication

- Install djangorestframework-simplejwt
- Configure JWT settings
- TokenObtainPairView untuk login
- TokenRefreshView
- Protected endpoints dengan JWT
- Custom claims
- Teknologi: JWT authentication

Hari 2: Custom User Model

Topik: AbstractBaseUser, BaseUserManager, custom authentication

Proyek: CustomUserAPI

Custom user dengan email login

- Create custom User model (email as username)
- BaseUserManager untuk create_user
- Update AUTH_USER_MODEL
- Migrations dari awal
- Serializer untuk custom user
- Teknologi: Custom User model

Hari 3: OAuth2 & Social Authentication

Topik: django-allauth, social login, OAuth2 flow

Proyek: SocialAuthAPI

Social login integration

- Install django-allauth

- Configure Google OAuth (simulation)
- Social authentication endpoints
- Link social accounts
- User data dari social providers
- Teknologi: OAuth2, django-allauth

Hari 4: API Security Best Practices

Topik: CORS, CSRF, rate limiting, security headers

Proyek: SecureAPI

Secure API implementation

- Install django-cors-headers
- Configure CORS_ALLOWED_ORIGINS
- Rate limiting dengan django-ratelimit
- Security headers
- Input sanitization
- SQL injection prevention
- Teknologi: API security

Hari 5: Password Policies & Security

Topik: Password validators, password strength, 2FA basics

Proyek: SecureAuthSystem

Enhanced authentication security

- Password validators configuration
- Password strength checker
- Account lockout after failed attempts
- Password reset with expiry
- Email verification (simulate)
- Teknologi: Authentication security

Hari 6: API Versioning

Topik: URL versioning, header versioning, namespace versioning

Proyek: VersionedAPI

API dengan versioning

- URL path versioning (v1, v2)
- Different serializers per version
- Backward compatibility
- Deprecation notices
- Version-specific logic
- Teknologi: API versioning

Hari 7: Review & Mini Project

Mini Project: SecureBankingAPI

Secure banking API simulation

Features:

- JWT authentication dengan refresh tokens
- Custom User model (email login)
- Account model (user, balance, account_number)
- Transaction model (from_account, to_account, amount)
- Transfer endpoint dengan validation
- Transaction history
- Rate limiting (5 transfers per minute)
- 2FA simulation dengan OTP
- Audit logs
- API versioning (v1)
- Comprehensive security measures
- Teknologi: JWT, Custom User, Rate Limiting, Security, Versioning

Minggu 2: File Handling & Media Management

Hari 8: Advanced File Uploads

Topik: File validation, image processing, file size limits

Proyek: FileUploadAPI

Secure file upload system

- File type validation (images only)
- File size limit (max 5MB)
- Custom upload_to path
- Filename sanitization
- Multiple file upload
- Teknologi: File upload validation

Hari 9: Image Processing

Topik: Pillow, image resize, thumbnails, image optimization

Proyek: ImageProcessorAPI

Image processing service

- Install Pillow
- Resize images on upload
- Generate thumbnails
- Image format conversion
- Compress images
- Different sizes untuk responsive
- Teknologi: Pillow, image processing

Hari 10: Cloud Storage (AWS S3 Simulation)

Topik: django-storages, S3 backend, boto3 basics

Proyek: CloudStorageAPI

Cloud storage integration

- Install django-storages
- Configure storage backends
- Simulate S3 upload (local)
- Separate media storage
- Static files on CDN (simulate)
- Teknologi: django-storages

Hari 11: File Download & Streaming

Topik: FileResponse, StreamingHttpResponse, download headers

Proyek: FileDownloadAPI

File download service

- Secure file download endpoint
- FileResponse untuk small files
- StreamingHttpResponse untuk large files
- Content-Disposition headers
- Access control (owner only)
- Download count tracking
- Teknologi: File downloads

Hari 12: CSV & Excel Export

Topik: Export QuerySet to CSV, openpyxl for Excel

Proyek: DataExportAPI

Export data to files

- Export products ke CSV
- Export orders ke Excel (openpyxl)
- Custom formatting
- Dynamic columns
- Filters before export
- Teknologi: CSV, openpyxl

Hari 13: PDF Generation

Topik: ReportLab, WeasyPrint, HTML to PDF

Proyek: PDFGeneratorAPI

Generate PDF reports

- Install ReportLab
- Generate invoice PDF
- Dynamic content
- Tables & styling
- Download endpoint
- Teknologi: ReportLab, PDF generation

Hari 14: Review & Mini Project

Mini Project: DocumentManagementAPI

Sistem manajemen dokumen

Features:

- Upload documents (PDF, images, Excel, CSV)
- File validation & virus scan simulation
- Folder/directory structure
- Metadata (title, description, tags)
- File versioning
- Search documents by metadata
- Generate thumbnails untuk images
- PDF preview
- Export document list ke Excel
- Download with access control
- Storage quota per user
- Audit trail (who uploaded, when, who downloaded)
- Teknologi: File handling, Image processing, PDF, Excel, Storage backends

Minggu 3: Background Tasks & Celery

Hari 15: Celery Setup & Basics

Topik: Celery installation, message broker, workers

Proyek: CelerySetup

Setup Celery dengan Django

- Install Celery
- Redis as message broker (simulate with RabbitMQ)
- celery.py configuration
- Start Celery worker
- Simple task example
- Teknologi: Celery basics

Hari 16: Async Tasks

Topik: @shared_task, delay(), apply_async()

Proyek: AsyncEmailSender

Asynchronous email sending

- Email sending task
- delay() untuk run async
- apply_async() dengan countdown
- Task status tracking
- Task result backend
- Teknologi: Celery tasks

Hari 17: Periodic Tasks

Topik: Celery Beat, crontab, periodic schedules

Proyek: ScheduledReports

Scheduled background jobs

- Install django-celery-beat
- Configure periodic tasks
- Daily report generation
- Cleanup old files weekly
- Crontab schedules
- Teknologi: Celery Beat

Hari 18: Task Monitoring & Error Handling

Topik: Flower, task retry, error handling

Proyek: TaskMonitoring

Monitor Celery tasks

- Install Flower
- Monitor task execution
- Retry failed tasks
- Max retries configuration
- Error logging

- Task timeout
- Teknologi: Flower, task monitoring

Hari 19: Complex Task Workflows

Topik: Task chains, groups, chords

Proyek: WorkflowTasks

Complex task orchestration

- Chain tasks sequentially
- Group parallel tasks
- Chord (group + callback)
- Task dependencies
- Workflow example: process order
- Teknologi: Celery workflows

Hari 20: WebSockets & Real-time (Introduction)

Topik: Django Channels basics, WebSocket concepts

Proyek: RealtimeNotifications

Real-time notifications intro

- Install channels
- ASGI configuration
- Simple WebSocket consumer
- Send notifications to connected clients
- Chat room basic example
- Teknologi: Django Channels basics

Hari 21: Review & Mini Project

Mini Project: VideoProcessingAPI

Video processing service dengan Celery

Features:

- Upload video endpoint
- Background video processing (simulate transcoding)
- Generate thumbnails dari video

- Multiple quality versions
- Progress tracking dengan Celery task
- WebSocket notification ketika selesai
- Queue management
- Retry failed processing
- Scheduled cleanup old videos
- Storage management
- API endpoint untuk task status
- Cancel processing task
- Teknologi: Celery, Celery Beat, Channels, File processing

Minggu 4: API Documentation & Production Prep

Hari 22: API Documentation - drf-spectacular

Topik: OpenAPI, Swagger, drf-spectacular

Proyek: DocumentedAPI

Auto-generated API docs

- Install drf-spectacular
- Configure spectacular
- `@extend_schema` decorator
- Customize endpoint docs
- Swagger UI
- ReDoc UI
- Teknologi: drf-spectacular, OpenAPI

Hari 23: Custom API Documentation

Topik: Custom schemas, examples, request/response docs

Proyek: EnhancedAPIDocs

Comprehensive API documentation

- Request examples
- Response examples
- Error response documentation
- Authentication documentation

- Endpoint descriptions
- Parameter descriptions
- Teknologi: API documentation

Hari 24: Logging & Monitoring

Topik: Python logging, log levels, log handlers, Sentry integration

Proyek: ProductionLogging

Production-ready logging

- Configure logging settings
- Different log levels per environment
- File handlers & console handlers
- Log rotation
- Sentry for error tracking (simulate)
- Custom log formatters
- Teknologi: Logging, monitoring

Hari 25: Environment Configuration

Topik: django-environ, .env files, environment variables

Proyek: EnvironmentConfig

Multi-environment configuration

- Install django-environ
- Separate settings (dev, staging, prod)
- Environment variables
- Secret management
- Database configs per env
- Debug settings
- Teknologi: django-environ

Hari 26: Database Migrations Best Practices

Topik: Migration strategies, data migrations, squashing

Proyek: MigrationManagement

Advanced migration handling

- Data migrations
- RunPython operations
- Reverse migrations
- Squash migrations
- Migration dependencies
- Zero-downtime migrations
- Teknologi: Django migrations advanced

Hari 27: Performance Testing & Profiling

Topik: django-silk, load testing, profiling queries

Proyek: PerformanceTesting

Test & optimize performance

- Install django-silk
- Profile slow endpoints
- Identify bottlenecks
- Load testing dengan locust (basic)
- Optimize queries
- Response time monitoring
- Teknologi: django-silk, performance testing

Hari 28: Review & Final Project Bulan 5

Final Project: SocialMediaAPI

Social media platform API

Models:

- CustomUser (email login)
- Profile (bio, avatar, followers)
- Post (content, image, video, author)
- Comment (post, author, content)
- Like (user, post)
- Notification (user, type, content, read)

Features:

- JWT authentication
- User registration & email verification
- Follow/unfollow users
- Create post (text, image, video)
- Video upload dengan background processing (Celery)
- Like/unlike posts
- Comment on posts
- Real-time notifications (WebSocket)
- News feed (posts from followed users)
- Search users & posts
- File upload dengan cloud storage (simulate)
- Image thumbnails generation
- Export user data ke PDF
- Rate limiting
- API documentation (Swagger)
- Comprehensive logging
- Multiple environments configuration
- Query optimization
- Caching popular posts
- Background cleanup tasks
- Teknologi: JWT, Celery, Channels, File processing, drf-spectacular, Logging, Caching

Progress Check Bulan 5:  JWT Authentication  Custom User model  OAuth2 & social auth  API security (CORS, rate limiting)  API versioning  Advanced file handling  Image processing  Cloud storage integration  PDF & Excel generation  Celery & background tasks  Periodic tasks (Celery Beat)  WebSockets basics (Channels)  API documentation (OpenAPI)  Production logging & monitoring  Environment configuration  Performance testing

Total Proyek: 60+ proyek & 4 major projects **Siap lanjut:** Bulan 6 - Deployment & DevOps